

E032

BSCNE-Cisco Capstone Project

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Two switches are connected by a single cable. VLANs 10 and 20 exist on both, but only VLAN 10 hosts can reach each other across the link. The cable is Cat6 and 15 meters long. Both switch ports are configured as access ports in VLAN 10. What is the fix?

- A. Replace the cable with a shorter one.
- B. Move both ports to VLAN 20.
- C. Add a router between the switches.
- D. Configure both ends as 802.1Q trunks allowing VLANs 10 and 20.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A switch port connects to a single desktop PC in VLAN 30. How should the port be configured?

- A. As a trunk with VLAN 30 as native.
- B. As a trunk carrying all VLANs.
- C. Shut down, since PCs never need switch ports.
- D. As an access port in VLAN 30.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

On a Layer 3 switch, you create SVIs for VLAN 10 (192.168.10.1) and VLAN 20 (192.168.20.1). Hosts use those addresses as their default gateways. The switch has 48 PoE ports. Hosts still cannot reach the other VLAN. What is the most likely missing step?

- A. VLANs can never communicate.
- B. The PoE budget is exceeded.
- C. IP routing is not enabled on the Layer 3 switch.
- D. The hosts need gateways in the other VLAN.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

You must stop host 10.1.5.20 from reaching the server subnet 10.1.9.0/24 while still letting it reach everything else. Only a standard ACL is allowed for this task. The host is a Windows 11 laptop in accounting. Where should the ACL be applied?

- A. Outbound on the router interface facing the server subnet (near the destination).
- B. On the accounting VLAN only, because of the OS.
- C. Anywhere, because placement does not matter.
- D. Inbound on the interface facing the host (near the source).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An extended ACL blocks only HTTP from subnet 10.2.3.0/24 to one web server. Where is it best applied?

- A. It does not matter since both types behave the same.
- B. Near the destination, like a standard ACL.
- C. Inbound on the interface closest to the 10.2.3.0/24 source.
- D. On every interface in the network.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An ACL entry must match subnet 192.168.10.64/27. The router is a branch model running a recent IOS release. Which wildcard mask is correct?

- A. 0.0.0.31
- B. 255.255.255.224
- C. 0.0.0.64
- D. 0.0.0.32

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An ACL uses wildcard mask 0.0.0.15. What prefix length does it match?

- A. /4
- B. /28
- C. /29
- D. /15

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A router has two routes to 172.16.40.0/24: one learned by OSPF and one static route added by a senior engineer during a migration. The OSPF route has a lower metric than its neighbors. Which route is installed?

- A. The static route (AD 1).
- B. The OSPF route, because of its low metric.
- C. Both, load-balanced.
- D. The OSPF route, since higher AD is preferred.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A router has an OSPF route to 10.1.1.0/24 and a static route to 10.1.0.0/16. Which route does it use for a packet to 10.1.1.5?

- A. Neither; the packet is dropped.
- B. The OSPF /24 route.
- C. The default route.
- D. The static /16 route, because static AD is lower.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

You finish an OSPF task, and the evaluator will log in tomorrow. The lab router sits on a shelf next to a UPS that was tested last month. Overnight the lab is rebooted for maintenance. What must you have done for your work to be graded?

- A. Tested OSPF neighbors.
- B. Saved the running configuration to the startup configuration.
- C. Nothing; the UPS keeps the configuration.
- D. Rebooted the router yourself to test it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A task requires the same VLAN and NTP settings on 40 switches. A teammate proposes a script that logs in by Telnet, types CLI lines, and scrapes output with text matching. The switches were bought in two batches. Which approach is better?

- A.** Configure each switch by hand.
- B.** Keep the Telnet script but run it twice to be sure.
- C.** An idempotent Ansible playbook in version control, or NETCONF/RESTCONF with YANG models.
- D.** Split the work by purchase batch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A branch router allows administrators to connect by Telnet from anywhere on the internal network. The branch has 25 employees and a single ISP. Which change best hardens management access?

- A.** Enable SSH but leave Telnet on as a backup.
- B.** Keep Telnet, since it is internal only.
- C.** Add a second ISP.
- D.** Disable Telnet, enable SSH with strong authentication, and restrict management access to an admin subnet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

Two switches are connected by a single cable. VLANs 10 and 20 exist on both, but only VLAN 10 hosts can reach each other across the link. ~~The cable is Cat6 and 15 meters long.~~ Both switch ports are configured as access ports in VLAN 10. What is the fix?

Target word: why only one VLAN crosses **Decides:** An access port carries one VLAN.

A	Camouflage	Cable length is within spec and does not decide.
B	Inverted operation	Swaps which VLAN fails.
C	Adjacent category	Routing is for between VLANs, not carrying VLANs across a link.
D	Correct	Trunks carry multiple tagged VLANs.

2. Answer D Concept 1 • transfer

A switch port connects to a single desktop PC in VLAN 30. How should the port be configured?

Target word: reverse: end-device port **Decides:** One VLAN to one end device.

A	Confusion set	Still a trunk; unnecessary for one device.
B	Inverted operation	Exposes other VLANs to the PC.
C	Absolute language	PCs connect through access ports.
D	Correct	End devices use untagged access ports.

3. Answer C Concept 2

On a Layer 3 switch, you create SVIs for VLAN 10 (192.168.10.1) and VLAN 20 (192.168.20.1). Hosts use those addresses as their default gateways. ~~The switch has 48 PoE ports.~~ Hosts still cannot reach the other VLAN. What is the most likely missing step?

Target word: missing piece for inter-VLAN routing **Decides:** SVIs and gateways are correct; routing must be enabled.

A	Absolute language	They can through Layer 3.
B	Camouflage	Power does not decide routing.
C	Correct	Without it, SVIs do not route between VLANs.
D	Inverted operation	Gateways must be in their own VLAN.

4. Answer A Concept 3

You must stop host 10.1.5.20 from reaching the server subnet 10.1.9.0/24 while still letting it reach everything else. Only a standard ACL is allowed for this task. ~~The host is a Windows 11 laptop in accounting.~~ Where should the ACL be applied?

Target word: standard ACL placement **Decides:** Standard ACLs match source only.

A	Correct	Near the source it would block the host from everything.
B	Camouflage	OS does not decide placement.
C	Absolute language	Placement changes what is blocked.
D	Inverted operation	That is extended ACL placement; a standard ACL here blocks all the host's traffic.

5. Answer C Concept 3 • transfer

An extended ACL blocks only HTTP from subnet 10.2.3.0/24 to one web server. Where is it best applied?

Target word: reverse: extended placement **Decides:** Extended ACLs can be specific, so place near the source.

A	Confusion set	Standard and extended ACLs match different fields.
B	Inverted operation	Wastes bandwidth carrying traffic that will be dropped.
C	Correct	Drops unwanted traffic early without over-blocking.
D	Adjacent category	Unneeded and error-prone.

6. Answer A Concept 4

An ACL entry must match subnet 192.168.10.64/27. ~~The router is a branch model running a recent IOS release.~~ Which wildcard mask is correct?

Target word: wildcard for /27 **Decides:** 255.255.255.255 – 255.255.255.224.

A	Correct	Inverse of 255.255.255.224.
B	Confusion set	That is the subnet mask.
C	Wrong stem number	Uses the network's last octet.
D	Dropped element	Block size without subtracting 1.

7. Answer B Concept 4 • transfer

An ACL uses wildcard mask 0.0.0.15. What prefix length does it match?

Target word: solve for prefix **Decides:** Block of 16 addresses = 4 host bits.

A	Inverted operation	Counts host bits instead of network bits.
B	Correct	$15 + 1 = 16$ addresses; $32 - 4 = 28$.
C	Dropped element	Forgets that the block size is the wildcard plus 1.
D	Wrong stem number	Reads the octet value as a prefix.

8. Answer A Concept 5

A router has two routes to 172.16.40.0/24: one learned by OSPF and one static route ~~added by a senior engineer during a migration. The OSPF route has a lower metric than its neighbors.~~ Which route is installed?

Target word: tie-break for the same prefix **Decides:** Same prefix → lower administrative distance.

A	Correct	Static AD 1 beats OSPF AD 110.
B	Camouflage	Metrics compare routes within one protocol, not across sources.
C	Adjacent category	Equal-cost sharing applies within one source.
D	Inverted operation	Lower AD is preferred.

9. Answer B Concept 5 • transfer

A router has an OSPF route to 10.1.1.0/24 and a static route to 10.1.0.0/16. Which route does it use for a packet to 10.1.1.5?

Target word: different prefixes **Decides:** Longest prefix match comes first.

A	Absolute language	Two valid routes match.
B	Correct	The /24 is the longer match; AD is not consulted.
C	Adjacent category	Used only when nothing more specific matches.
D	Inverted operation	AD only breaks ties for the same prefix.

10. Answer B Concept 6

You finish an OSPF task, and the evaluator will log in tomorrow. The lab router sits on a shelf next to a UPS ~~that was tested last month~~. Overnight the lab is rebooted for maintenance. What must you have done for your work to be graded?

Target word: persist configuration **Decides:** Only saved startup configuration survives reload.

A	Dropped element	Good practice, but drops the save.
B	Correct	Otherwise changes are lost on reload.
C	Camouflage	Reboot clears unsaved changes regardless of power backup.
D	Inverted operation	Without saving, this erases the work.

11. Answer C Concept 7

A task requires the same VLAN and NTP settings on 40 switches. A teammate proposes a script that logs in by Telnet, types CLI lines, and scrapes output with text matching. ~~The switches were bought in two batches.~~ Which approach is better?

Target word: automation approach **Decides:** Fragile, non-idempotent screen scraping vs. structured tools.

A	Inverted operation	Slow and error-prone at 40 devices.
B	Dropped element	Not idempotent and insecure.
C	Correct	Structured, repeatable, safe to re-run.
D	Camouflage	Purchase history does not decide the method.

12. Answer D Concept 8

A branch router allows administrators to connect by Telnet from anywhere on the internal network. ~~The branch has 25 employees and a single ISP.~~ Which change best hardens management access?

Target word: management-plane hardening **Decides:** Telnet sends credentials in clear text.

A	Dropped element	Leaves the insecure path open.
B	Absolute language	Internal networks are not automatically safe.
C	Camouflage	Redundancy does not harden access.
D	Correct	Encrypted, authenticated, limited access.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Access vs. trunk ports	An access port carries one VLAN untagged to an end device; a trunk carries multiple VLANs between switches using 802.1Q tags (the native VLAN is untagged).	Linking two switches with an access port and expecting all VLANs to pass.	A switch-to-switch link where only one VLAN works.
2	Inter-VLAN routing	Hosts in different VLANs need a Layer 3 path — router subinterfaces on a trunk (router-on-a-stick) or SVIs on a Layer 3 switch with routing enabled — and each host's default gateway must point to it.	Creating SVIs but leaving IP routing disabled or gateways wrong.	Hosts reach their own VLAN but not others.
3	ACL type and placement	Standard ACLs match only source address and go near the destination; extended ACLs match source, destination, protocol, and port and go near the source. Every ACL ends with an implicit deny.	Placing a standard ACL near the source, which blocks the host from everything.	Which ACL type and which interface to apply it to.
4	Wildcard masks	A wildcard mask is the inverse of the subnet mask: 255.255.255.255 minus the mask (for a /27, 0.0.0.31).	Entering the subnet mask where a wildcard is required, or using the block size without subtracting 1.	An ACL or OSPF network statement for a subnet.
5	Route selection order	A router first picks the longest prefix match; administrative distance (connected 0, static 1, OSPF 110, RIP 120) breaks ties only between routes to the same prefix.	Letting a lower administrative distance beat a longer prefix.	Two routes that cover the same destination.
6	Running vs. startup configuration	Changes go to the running configuration immediately; they survive a reload only after being saved to the startup configuration.	Assuming working changes are automatically permanent.	Changes disappear after a reboot.
7	Network automation	Tools like Ansible are agentless and apply idempotent, version-controlled playbooks; model-driven interfaces (NETCONF/RESTCONF with YANG) give structured, programmatic configuration.	Scripting raw CLI screen-scraping that breaks when output changes, or re-running non-idempotent scripts.	Configuring many devices the same way.
8	Management-plane hardening	Use SSH (not Telnet), strong local or AAA authentication, restricted management access, and shut down unused ports.	Leaving Telnet enabled because it is "inside the network."	How administrators connect to devices.

Course scope (public description): In the BSCNE-Cisco Capstone Project, students choose from a fixed set of tasks that they might encounter in the workplace. Examples might include configuring network components, applying security policies to a set of cloud resources, or automating a network deployment using orchestration tools. Students will submit their tasks for evaluation in a working, deployed environment where an evaluator can check their results by inspecting configurations or logging into machines or services to test functionality. All tasks will be completed integrating with Cisco tools and technologies.